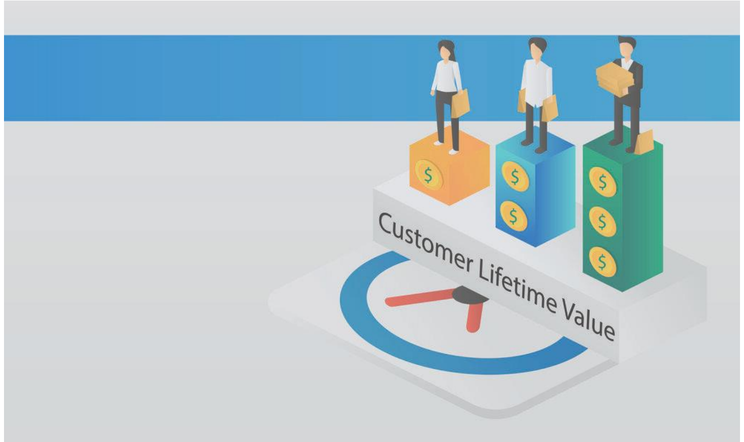
Machine Learning for Optimizing Customer Lifetime Value (CLTV) Prediction

Introduction

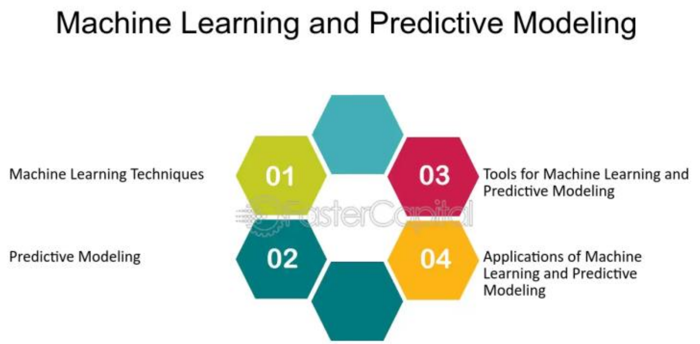
- Sub-study focuses on Customer Lifetime Value (CLTV) prediction
- Applies machine learning based on demographic and usage data.
- Emphasizes the value of CLTV for marketing and retention efforts.



Research Aim

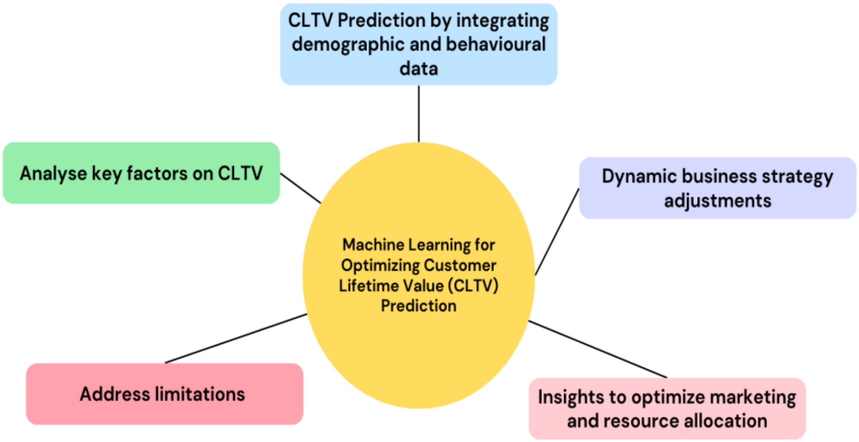
Goal:

To develop advanced machine learning models for accurately estimating Customer Lifetime Value (CLV) by incorporating demographic data, usage patterns, and behavioural insights to enhance predictive capabilities and support strategic business decisions.



Objectives

- Enhance Customer Lifetime Value (CLTV) prediction models by integrating demographic and behavioural data.
- Analyse key factors like geography, income, and product usage on Customer Lifetime Value.
- Address limitations of static models by incorporating real-time data.
- Provide actionable insights to optimize marketing and resource allocation.
- Develop practical tools for dynamic business strategy adjustments.

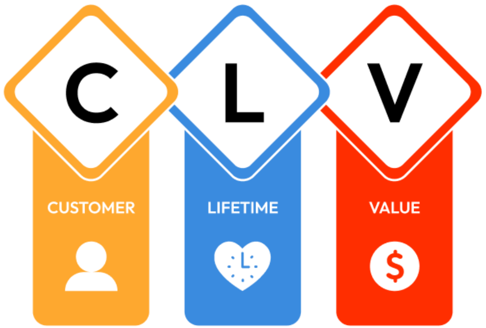


Novelty of the Study

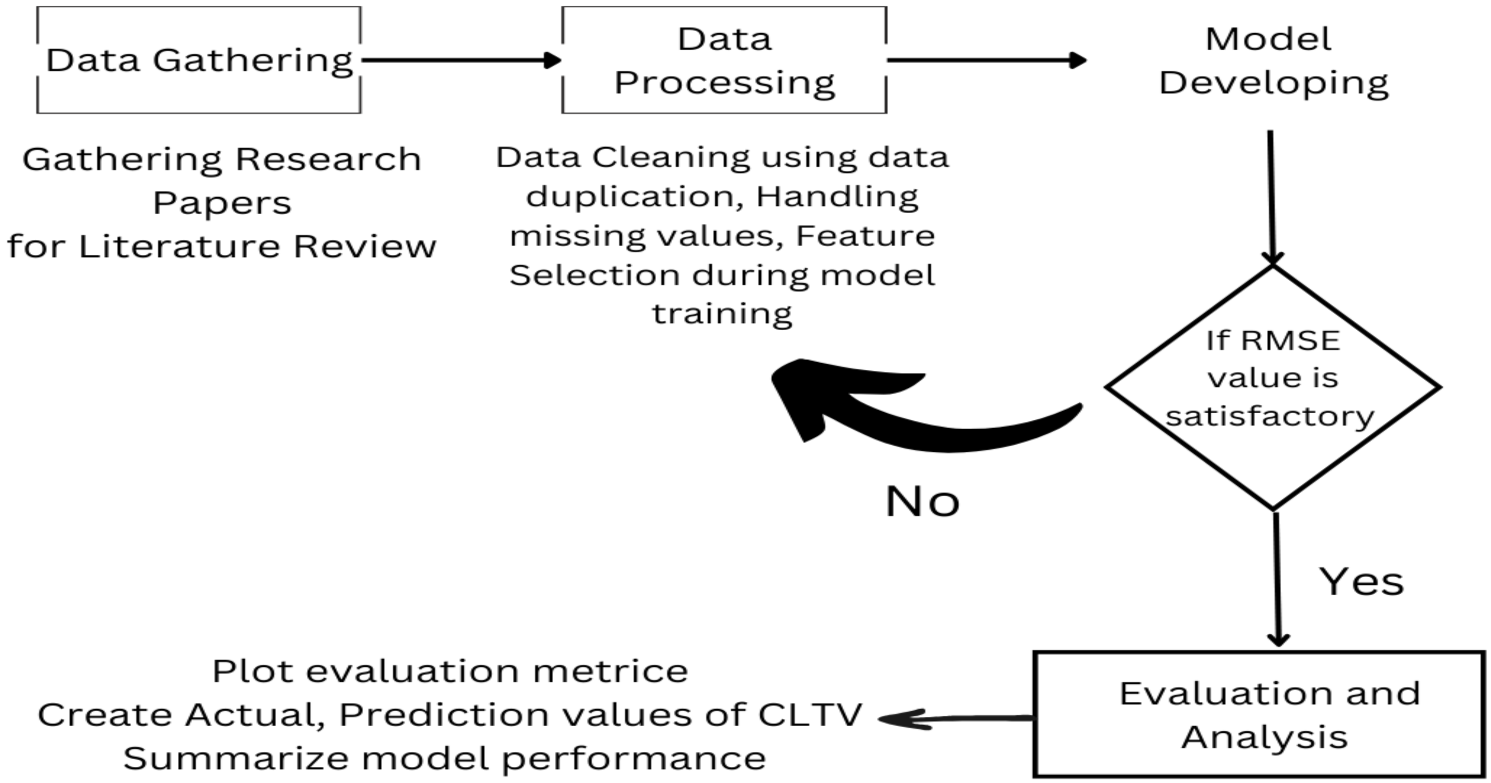
- Combines demographic and behavioral data with machine learning to improve Customer Lifetime Value (CLTV) predictions.
- Introduces XGBoost as the most effective model for CLTV forecasting.
- Resolves inconsistencies in traditional CLTV estimation methods with ensemble techniques.
- Provides actionable insights for businesses to enhance customer retention and optimize resources.
- Proposes future integration of real-time data for dynamic and adaptive predictions.

Problem Statement

- No behavioral data for CLTV methods currently.
- Challenges: Static Models, Outdated Data, Low Accuracy
- Solution: Advance ML with real time and demographic integration



Research Methodology



Algorithms Used

Linear Regression

Simple and interpretable, making it a good baseline model for predicting CLTV with linear relationships.

Random Forest

Combines multiple decision trees to reduce overfitting, providing robust predictions with improved accuracy and stability.

Decision Tree Regressor

Captures non-linear patterns and interactions in the data, offering flexibility for more complex relationships in CLTV prediction.

AdaBoost Regressor

Focuses on improving weak models, boosting performance by reducing bias and handling noisy data well in CLTV prediction.

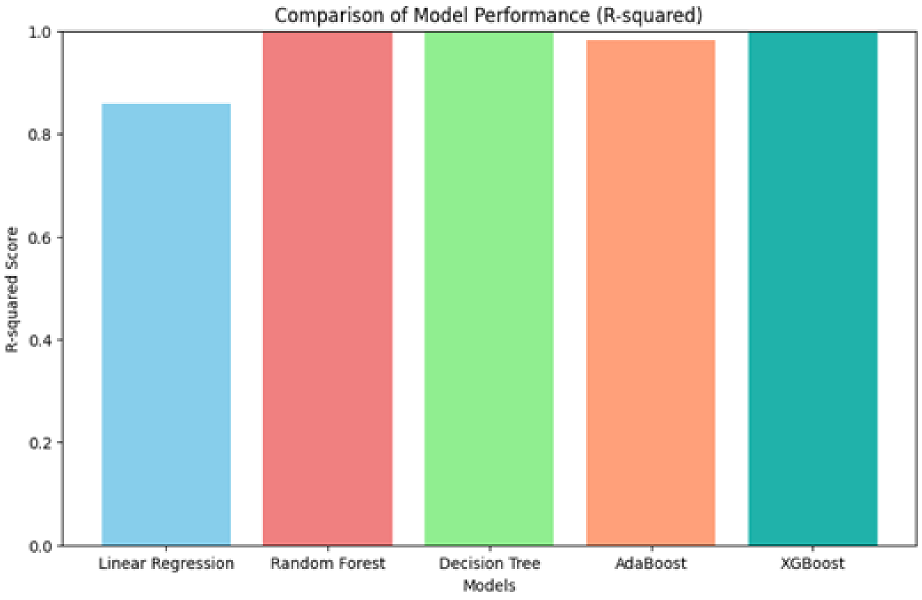
XGBoost Regressor

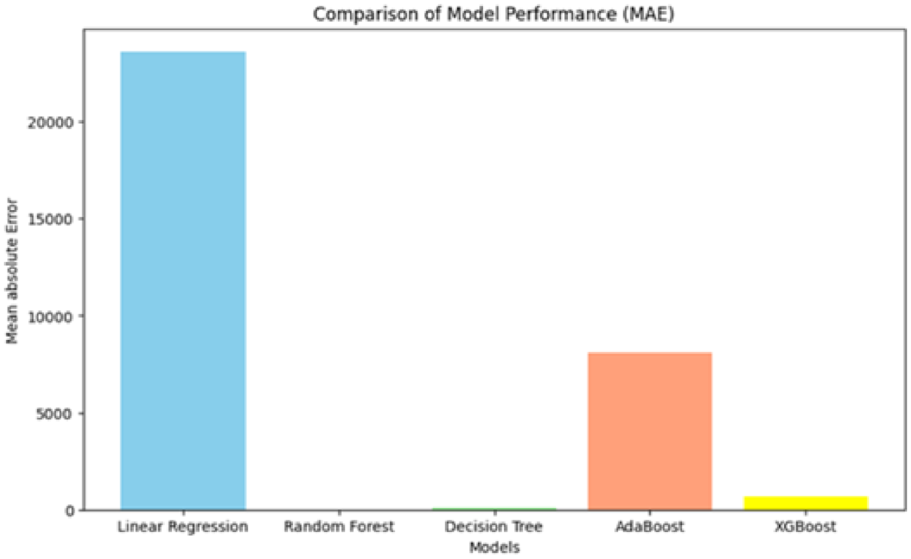
A highly efficient gradient boosting algorithm that optimizes both bias and variance, often delivering state-of-the-art performance for Customer Lifetime Value predictions.

Model Performance Comparison

Model	RMSE	R-Squared	MAE	MSE
Linear Regression	33732.76	0.85	23619.33	1137899557.54
Random Forest	50.10	0.99	5.31	2510.77
Decision Tree Regressor	306.40	0.99	95.52	93886.27
AdaBoost Regressor	12117.86	0.98	8116.48	146842709.33
XGBoost Regressor	3059.94	0.99	670.75	9363271.57

Comparison of Models





Business Implications

- **Benefits:** Better marketing, retention, allocation of resources.
- **Recommendation's:** Increase investment into ML, real-time data and expanding features.
- **Difficulties**: Type of data, quality of data, model interpretability.

Conclusion and Future Perspectives

Conclusion:

- ML improves CLTV estimation and decision-making.
- Traditional methods are surpassed by ensemble models.
- Actionable insights for maximizing the value of the customer.

Future:

- Real-time data, explainable AI, and sector-specific models.
- Then you should adopt ensemble models like XG Boost and Random Forest.
- Incorporate real-time data.
- Develop unified data systems.
- Optimize models continuously.

Goal:

Drive innovation in customer analytics.

Reference

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